

# TerraA I

By Phil Gannon, Geronimo Walker, Bryan  
***Detecting Illegal Cannabis Cultivations in  
Northern California National Forests***  
Burzon and Maria Lee



# TerraAI Team



**Maria Lee**



**Geronimo Walker**



**Bryan Burzon**



**Phil Gagnon**

# Background



**Destruction of  
Habitat**



**Wildfire Risk**



**Water Theft**



**Chemical  
Pollution / Hazard**



**Trash and Waste**

# Objectives and target audience



## Project objectives

- Detecting deforestation using forest cover with Machine Learning algorithm
- Using geospatial analysis to predict potential sites such as water proximity and land slope



## Goals for impact

- Detect potential illegal cannabis cultivation sites in the four national forests
- More efficient detection of environmental degradation

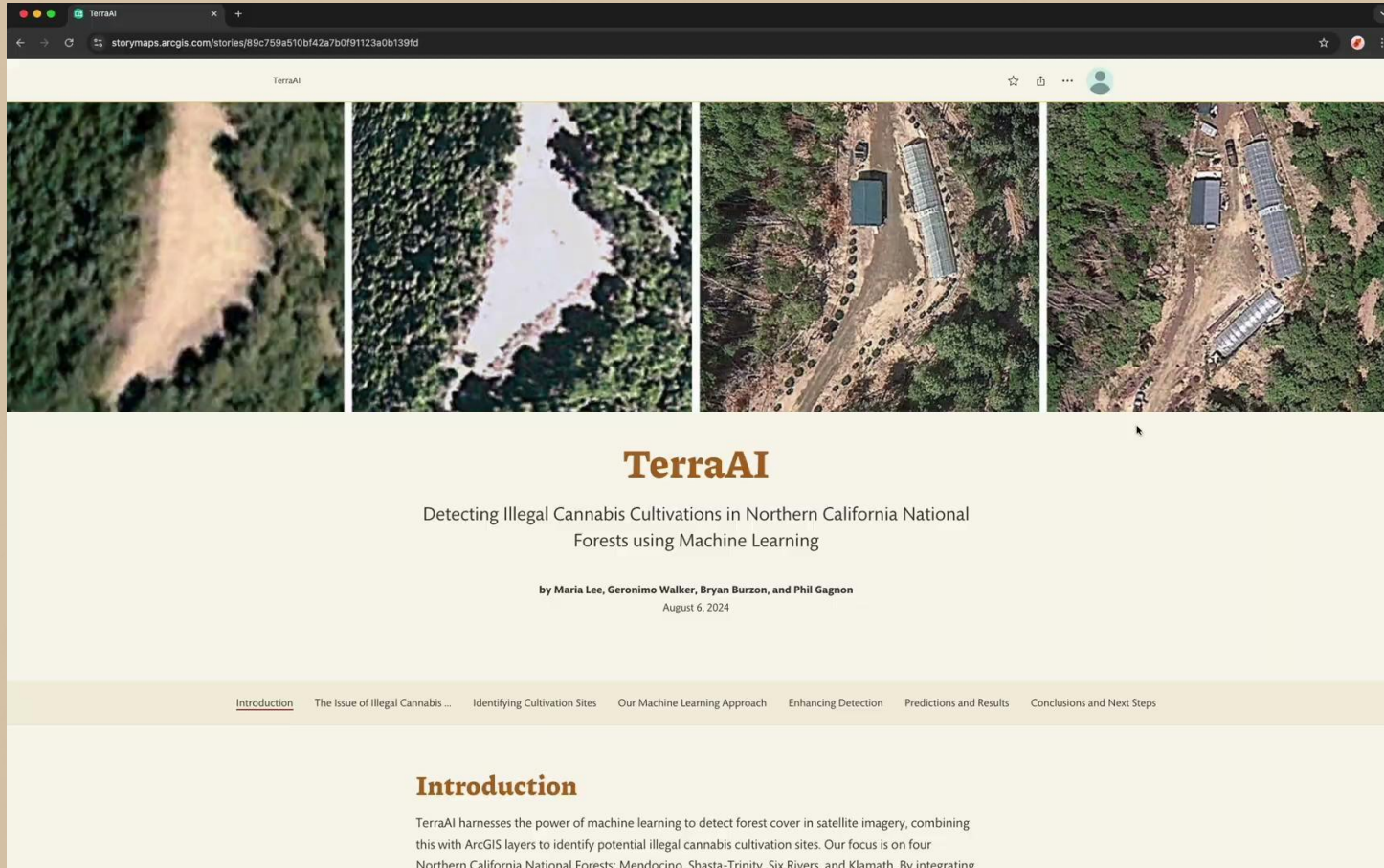


## Target audience

- USDA Forest Service Personnel
- California Water Resources Control Board
- Public

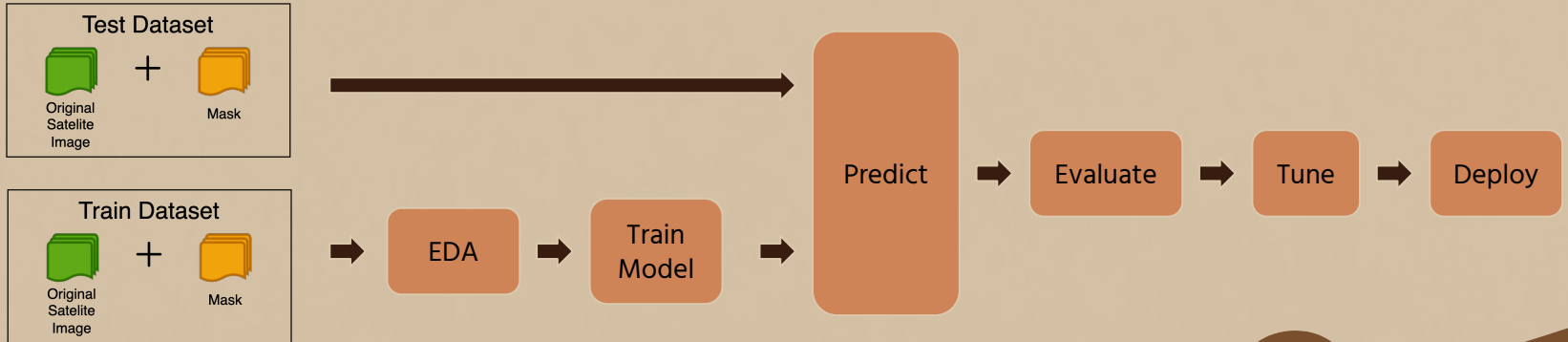
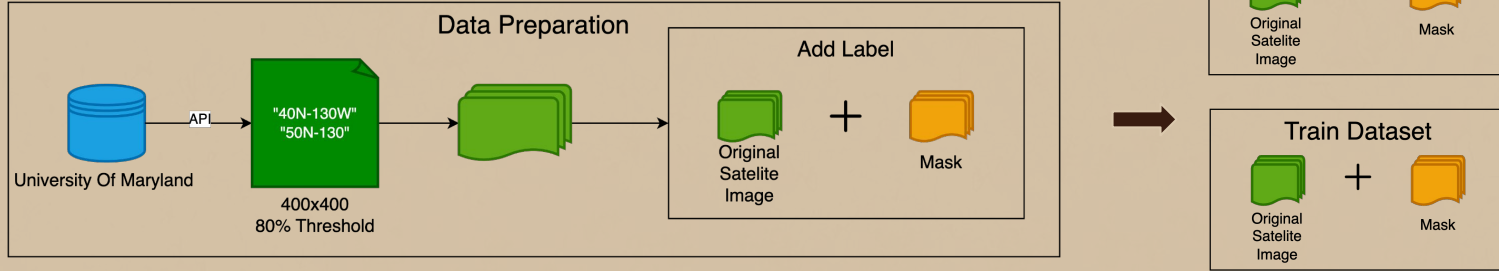


# Website DEMO

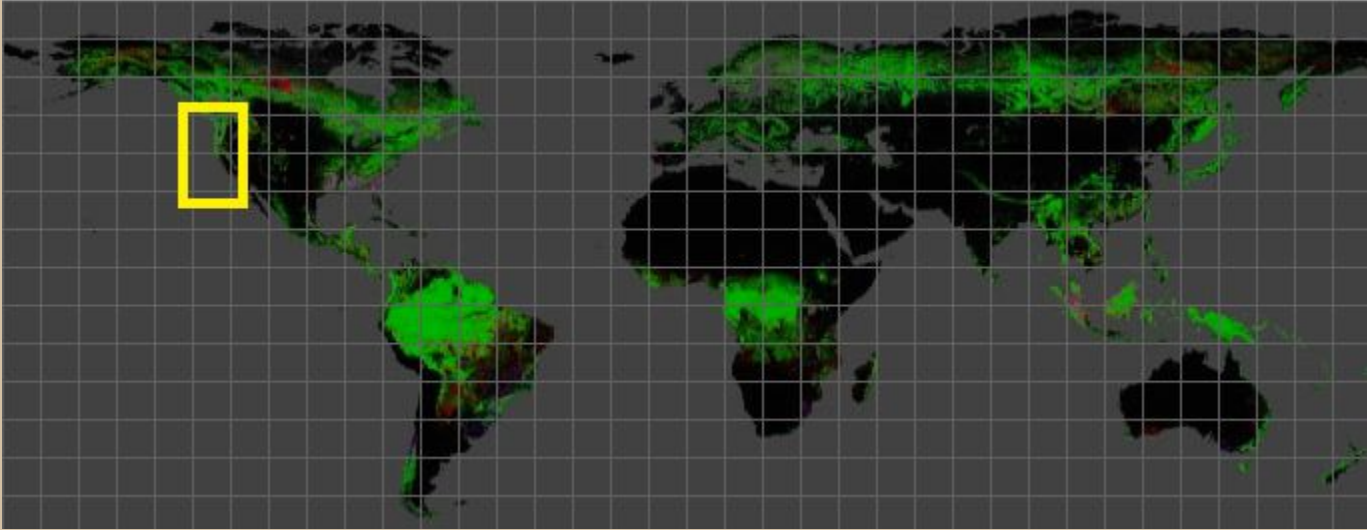


**TerraAI Story Maps Link: <https://storymaps.arcgis.com/stories/89c759a510bf42a7b0f91123a0b139fd>**

# Technical Pipelines



# EDA and Pre-processing



Source: Hansen/UMD/Google/USGS/NASA

# EDA and Pre-processing

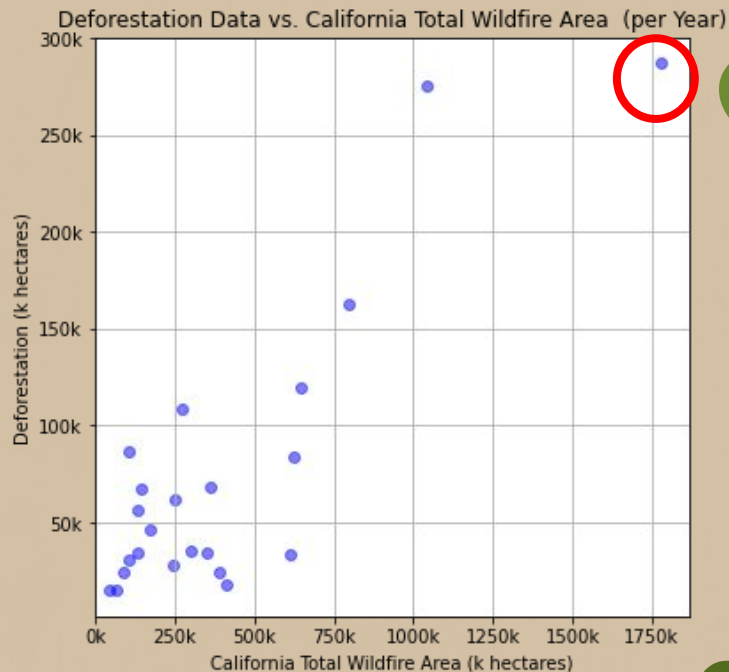
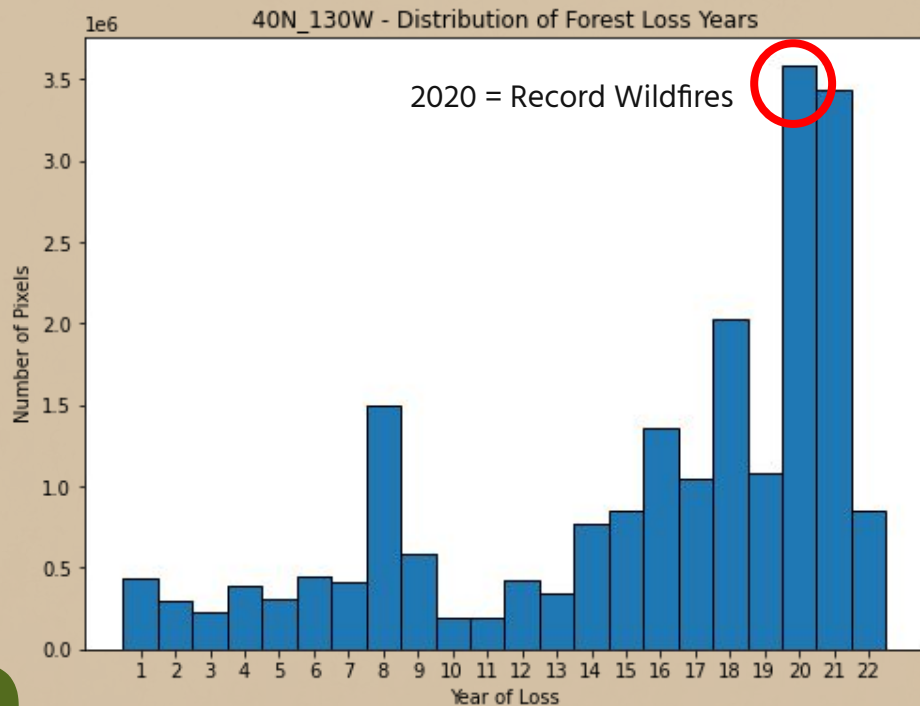
**Land surface in image (40N, 130W):**

**Trade-off between each image's dimensions (in pixel x pixel width), versus number of files**

| <b>Pixel width</b> | <b>Number of files</b> | <b>Latitude (km)</b> | <b>Longitude (km)</b> | <b>Area (km<sup>2</sup>)</b> |
|--------------------|------------------------|----------------------|-----------------------|------------------------------|
| 400                | 451                    | 11.10                | 8.62                  | 95.68                        |
| 200                | 1,804                  | 5.55                 | 4.31                  | 23.92                        |
| 100                | 7,216                  | 2.78                 | 2.16                  | 5.98                         |
| 50                 | 28,864                 | 1.39                 | 1.08                  | 1.50                         |
| 25                 | 115,456                | 0.69                 | 0.54                  | 0.37                         |



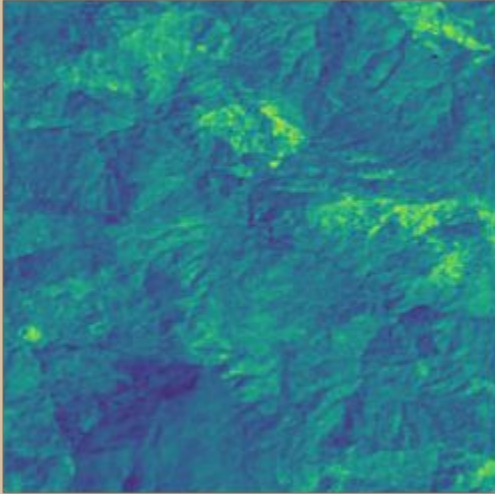
# EDA and Pre-processing



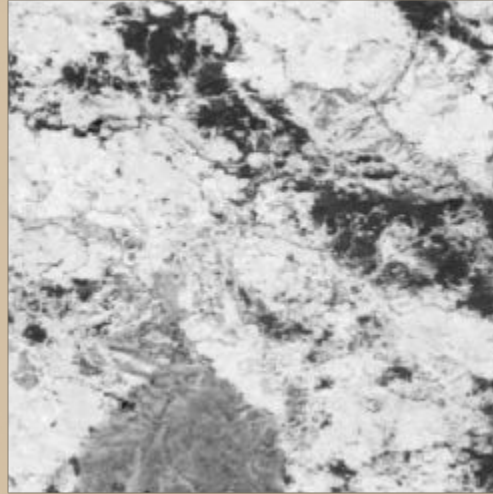
Source: National Interagency Fire Center

# EDA and Pre-processing

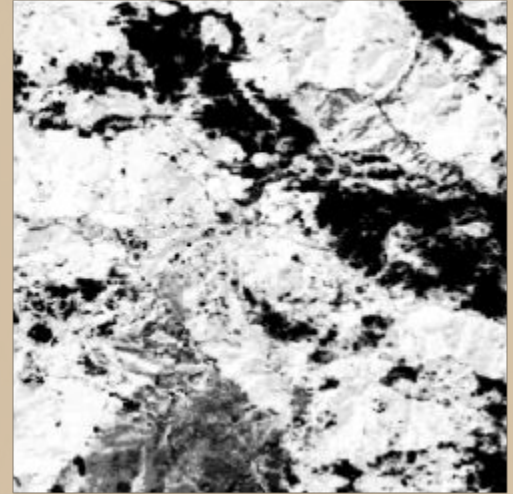
Raw Image (4 color channels)



NDVI Pre-processed Image



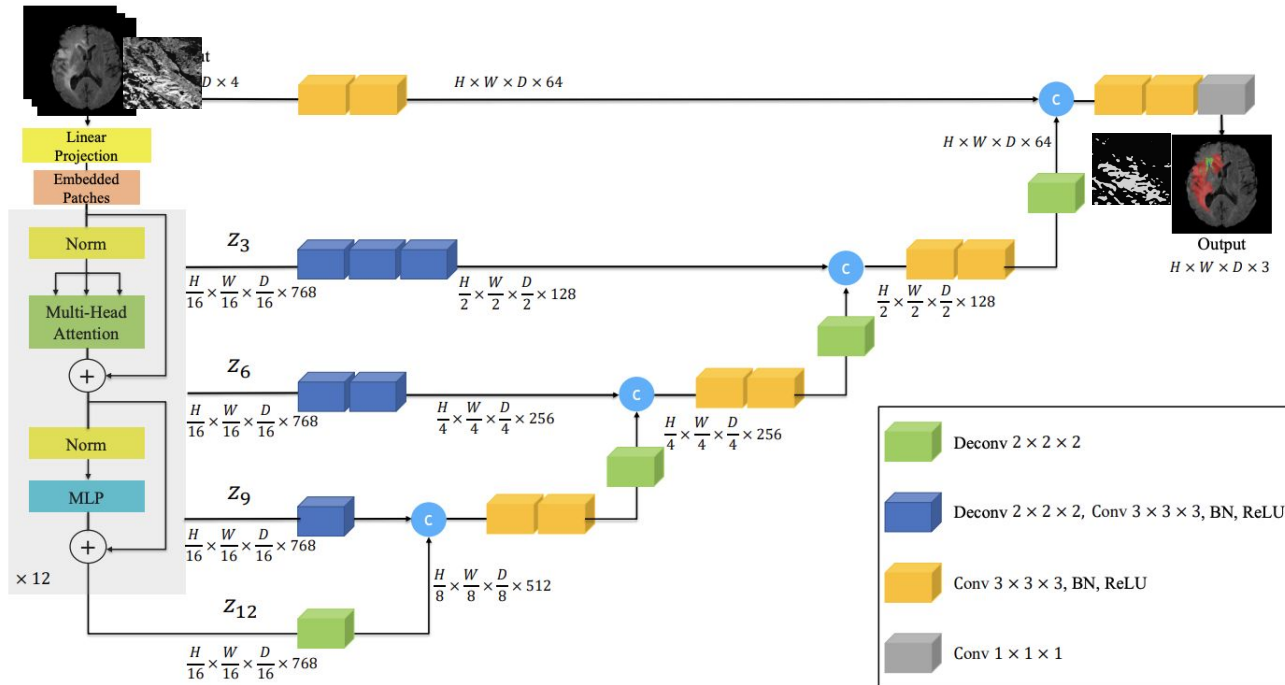
Tree cover labels



# Machine Learning Algorithm

## UNET Transformer Model

Transformer  
Encoder

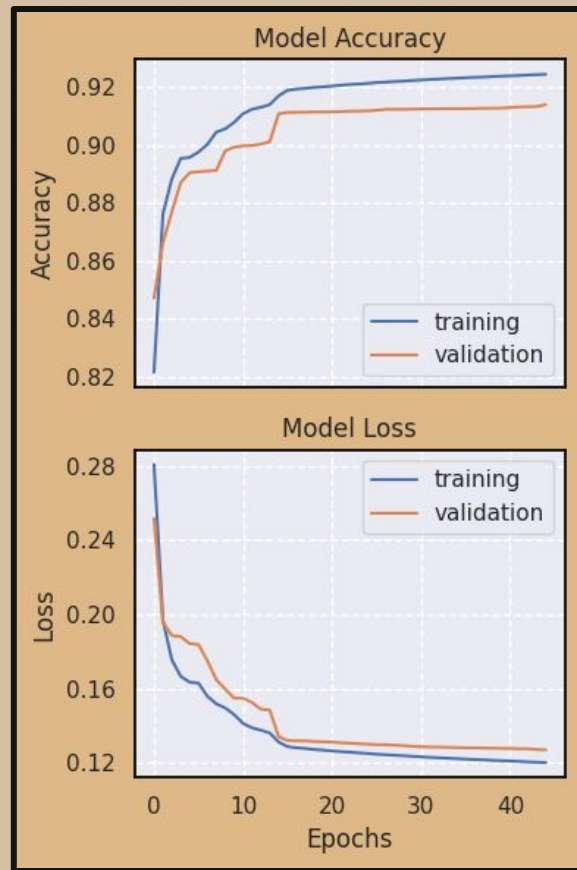


Cnn  
Decoder

# Model Evaluation

## Performance Metrics

|             |            |
|-------------|------------|
| ❖ Accuracy  | <b>91%</b> |
| ❖ Precision | <b>82%</b> |
| ❖ Recall    | <b>94%</b> |
| ❖ Dice Coef | <b>87%</b> |
| ❖ IoU       | <b>83%</b> |

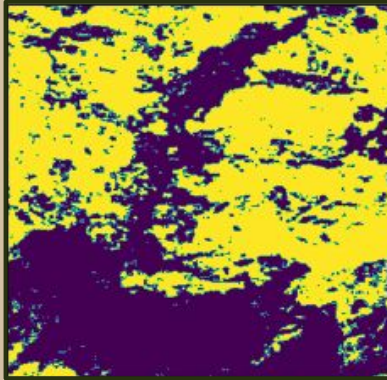
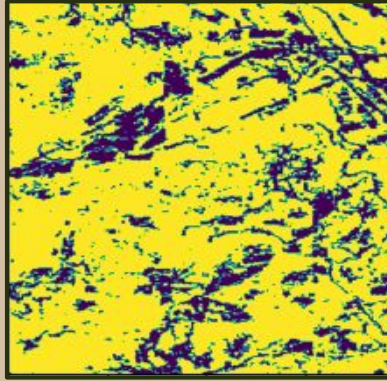


# Forest Cover Predictions

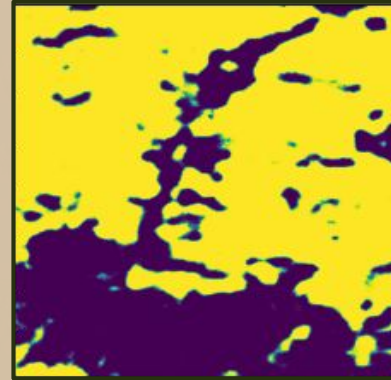
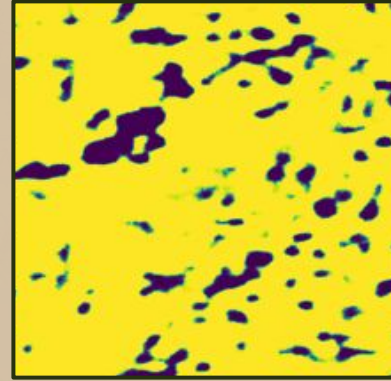
Original image



True Mask



Predicted Mask

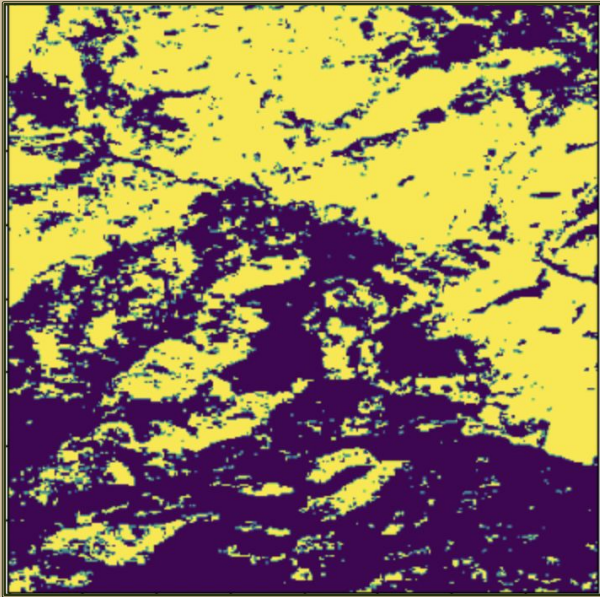




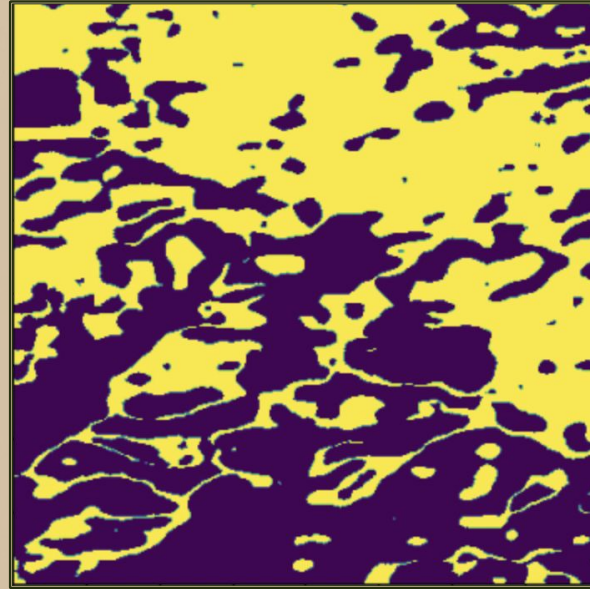
# Differencing Masks

Compare our calculated forest loss to a true forest loss provided by our training dataset

True Forest Loss



Predicted Forest Loss



# Deforestation Predictions

Apply our validated techniques to satellite images from copernicus

Earlier Image



Later Image



Predicted Deforestation



# Future Work



## Scale up

- Utilize Tech stack to its full potential
- More historical Images



## Obtaining Labeled Data

- USDA slow and protective
- Cannabis sites Labeled Data



**ArcGIS**

## Model Upgrades

- More Model variety
- More fine tuning

# Conclusion



Efficiency in Detection



Public Awareness



Goals Of Protecting National Forest



**Thanks!**